



Roll Number

NARULA INSTITUTE OF TECHNOLOGY
An Autonomous Institute under MAKAUT
2026
Even Semester - 2025-26
M201 - ENGINEERING MATHEMATICS - II

Time: 3 Hrs

Maximum Marks : 60

Instructions to the candidate:*Figures to the right indicate full marks.**Draw neat sketches and diagram wherever is necessary.**Candidates are required to give their answers in their own words as far as practicable***Part A****Answer any ten from the following, choosing the correct alternative of each question: 10×1=10**

1. The integrating factor of the differential equation $(x \log x) dy/dx + y = 2 \log x$ is (1) CO1 BL1 PO1
 - (a) $2 \log x$
 - (b) $x \log x$
 - (c) $\log x$
 - (d) $\log x^2$
2. if $\frac{dx}{dt} - y = 0$ and $\frac{dy}{dt} + x = 0$, then the solution of x is (1) CO2 BL1 PO1
 - (a) $A \cos t + B \sin t$
 - (b) $A \cos 2t + B \sin 2t$
 - (c) $A \cos t - B \sin t$
 - (d) $-A \cos 2t + B \sin 2t$
3. Find $L\{(1+t)^2\}$ (1) CO3 BL2 PO1
 - (a) $\frac{1}{s} + 2\frac{1}{s^2} + \frac{2!}{s^3}$
 - (b) $\frac{1}{s} + \frac{1}{s^2} + 2\frac{1}{s^3}$
 - (c) $\frac{1}{s^2} + \frac{2}{s} + \frac{1}{s^3}$
 - (d) none of the above
4. Runge Kutta formula has a truncation error which is of the order of (1) CO3 BL4 PO1
 - (a) h^2
 - (b) h^4
 - (c) h^5
 - (d) none of these
5. The general solution of $p = \log(px - y)$ where $p = \frac{dy}{dx}$ is (1) CO1 BL1 PO1
 - (a) $y = cx - c$.
 - (b) $y = cx - e^c$.
 - (c) $y = c^2x - e^{-c}$.
 - (d) None of these.

6. Which of the following correctly represents the number 0.007846 after (1) rounding off to three significant digits? C03 BL2 PO1
- (a) 0.00784. (b) 0.00785.
(c) 0.00780. (d) 0.00790.
7. The degree of precision of Simpson's one-third rule is (1) C03 BL1 PO1
- (a) 1. (b) 2.
(c) 3. (d) 4.
8. The order and degree of the differential equation $\left(\frac{d^2y}{dx^2}\right)^{\frac{1}{2}} + \frac{dy}{dx} + y = 0$ are respectively (1) C01 BL2 PO1
- (a) 2, 3. (b) 2, 4.
(c) 1, 2. (d) 2, 1.
9. $L^{-1}\left\{\frac{s}{s^2+a^2}\right\}$ is equal to (1) C02 BL3 PO1
- (a) $\frac{1}{2a}t\sin at.$ (b) $\frac{1}{a}t\sin at.$
(c) $\frac{1}{a^2}\sin at.$ (d) $\cos at.$
10. Newton's forward formula is used for interpolating the value of y near the (1) C01 BL1 PO1
- (a) Beginning of a set (b) end of a set
(c) middle of a set (d) none of these
11. The C.F. of the equation $(x^2D^2 - 2xD)y = 3x$ is (1) C01 BL1 PO1
- (a) $Ax + be^{3x},$ (b) $A + Be^{3x},$
(c) $Ae^x + Be^{2x},$ (d) $Ax + Be^{2x}.$
12. The value of $(1 + \Delta)(1 - \nabla)$ is (1) C01 BL1 PO1
- (a) 0 (b) 1
(c) 2 (d) 3

Part B

(Answer any four of the following) 5x4=20

13. Solve: $(D^2 - 2D)y = e^x \sin x$ (5) C02 BL3 PO2
14. Find $L(t e^{-2t} \sin 3t)$ (5) C03 BL1 PO1

15. Find missing term in the following tables: (5) C04 BL3 PO1
- x : 0 2 4 6 8
- f(x): 0 5 17 _ 65

16. Apply the convolution theorem to find the inverse Laplace transform of $F(s)=1/\{s(s+2)\}$. (5) C03 BL3 PO1

17. During a high-speed CNC machining operation, the vibration amplitude of the machine tool is monitored at different spindle speeds to assess machining stability and component quality. The recorded data are given below: (5) C03 BL3 PO1

Load Application Rate (Cycles / sec)	100	200	300	400	500
Response Time (ms)	50	120	240	410	630

Apply Newton's Backward Interpolation Method to estimate the vibration amplitude at a spindle speed of 450 rpm.

18. Find inverse laplace transform of $\frac{8e^{-3s}}{s^2 + 4}$. (5) C04 BL4 PO10

Part C

(Answer any three of the following) 3x10=30

19. In a cloud computing environment, the data processing response $y(t)$ of a distributed server subjected to a continuously increasing workload is governed by the following equation $\frac{d^2y}{dt^2} + 4y = te^t$, where t represents the operating time of the server.
- (a) Determine the data processing response $y(t)$ at any specified time t . (6) C01 BL3 PO1
- (b) Given the initial conditions $y(0) = 0$ and $y'(0) = 0$, analyze the long-term behavior of the data processing response as $t \rightarrow \infty$, and examine the stability of the server under continuously increasing workload conditions. (4) C04 BL4 PO2
20. (a) Solve: $(x^2D^2 - xD - 3)y = x^2\log x$ (5) C02 BL3 PO1
- (b) Using Convolution theorem verify that $\int_0^t \sin u \cos(t-u) du = \frac{t \sin t}{2}$ (5) C03 BL1 PO1

21. (a) find the value of the integral $\int_0^1 e^x dx$, by Trapezoidal rule with $h=0.1$ and correct upto four decimal places. (5) C04 BL2 PO2

- (b) Using approximate formula find $f(0.29)$ from the following table: (5) C04 BL2 PO1

x :	0.20	0.22	0.24	0.26	0.28
	0.30				
f(x):	1.6596	1.6698	1.6804	1.6912	1.7024
	1.7139				

22. (a) In a cloud-based database system, the query response times (in *milliseconds*) were recorded at different points during a load test as shown below: (5) C03 BL3 PO1

Time (Requests per Second)	100	300	600
Query Response Time (ms)	150	250	400

Estimate the query response time at 400 requests per second using Lagrange interpolation.

- (b) In a precision manufacturing unit, a vibration isolation system is installed to minimize unwanted oscillations of a machine. The displacement (in *cm*) $y(t)$ of the spring-mass-damper system at time t seconds is governed by $\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 6y = 0$ with initial conditions $y(0) = 2, y'(0) = -1$. Determine the displacement response $y(t)$ as a function of time using the Laplace Transform technique, and analyze the effectiveness of the vibration isolation system in suppressing vibrations and ensuring stable operation over time. (5) C04 BL4 PO2

23. (a) Find the general solution of $y = (p + p^2)x + p^{-1}$. (5) C02 BL2 PO2

- (b) Solve the differential equation using Laplace transform: $y'' - 3y' + 2y = 0$, where $y(0)=1$ and $y'(0)=-1$. (5) C03 BL3 PO2